

LITERATURE SURVEY

AI/ML-Based Object Detection System for Dual-Energy X-Ray Luggage Scanners

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1. Introduction

Airport security screening is a mission-critical operation that directly impacts public safety. At the core of this process are X-ray baggage scanner systems, which security officers use to visually inspect the contents of passenger luggage. However, the ever-increasing volume of air passengers, compounded by the cognitive fatigue of human operators and the growing sophistication of concealment techniques, has created an urgent demand for intelligent, automated threat detection systems.

The field of AI-based object detection in X-ray imagery has gained significant traction over the past decade. Researchers have explored a wide spectrum of methodologies — ranging from classical machine learning pipelines utilizing handcrafted features, to modern deep learning architectures including Convolutional Neural Networks (CNNs), YOLO-family detectors, and Transformer-based models. Dual-energy X-ray technology, which encodes material composition information through color-coded pseudo-images, has emerged as the preferred imaging substrate for such automated systems.

This literature survey reviews and synthesizes the most influential published research in the following interconnected domains: (i) traditional X-ray image processing and threat recognition, (ii) CNN-based classification of dual-energy X-ray scans, (iii) real-time object detection using YOLO architectures, (iv) material classification in X-ray imagery, (v) dataset construction and augmentation strategies, and (vi) emerging directions including property-based detection and scalable model architectures. The survey establishes a comprehensive theoretical and empirical foundation for the proposed two-model, property-based detection system.

2. Background: Dual-Energy X-Ray Technology in Security Screening

Dual-energy X-ray baggage inspection systems (XBIS) function by emitting two beams of X-radiation at different energy levels — typically referred to as High Energy (HE) and Low Energy (LE). The differential absorption of these beams by materials with varying atomic numbers and densities allows the system to estimate material composition. The resulting pseudo-colored images assign distinct hues to material categories: organic substances appear orange, inorganic metallic materials appear blue or dark, and mixed-composition objects appear green or grey [1].

Deshpande et al. (2022), from Bhabha Atomic Research Centre (BARC), provide a foundational treatment of this imaging modality in their study published in the BARC Newsletter. Their work demonstrates that processing a four-channel input — derived as combinations of HE and LE scans — significantly improves object classification accuracy compared to single-channel approaches. The four-channel representation (-log HE, -log LE, HE+LE, HE-LE) captures both material discrimination and occlusion characteristics, making it particularly well-suited for training deep learning models on baggage imagery [7].

A comprehensive review by Mery et al. (2020), published in IEEE Access, surveys the state of X-ray baggage inspection systems and identifies three primary research dimensions: multi-energy imaging, multi-view scanning geometry, and computer vision algorithms [11]. The review validates that dual-energy imaging is the de facto standard for airport baggage screening because of its superior material differentiation capability, which is foundational to the property-based detection approach explored in this project.

The European Commission Joint Research Centre (JRC) Policy Report (2022) further contextualizes the regulatory and technological trajectory of X-ray screening systems. The report discusses how dual-energy CT (DECT) is increasingly being deployed at airports due to its ability to produce three-dimensional images with enhanced material resolution, and how emerging AI systems are being integrated as automated threat detection add-ons to existing hardware infrastructure [11].

3. CNN-Based Classification for Threat Recognition

Convolutional Neural Networks have become the dominant paradigm for image classification in X-ray security applications. Their ability to automatically learn spatially hierarchical features from raw image

data, without requiring handcrafted descriptors, makes them particularly effective for the complex, cluttered visual environment of baggage scans.

3.1 VGGNet, AlexNet, and Ensemble Architectures

Deshpande et al. (2022) conducted a rigorous comparative evaluation of AlexNet and VGGNet architectures on dual-energy X-ray images of baggage. VGG-19, when trained using the Adam optimizer on four-channel dual-energy inputs, achieved a testing accuracy of 99.05% and an AUC-ROC of 0.9924. The study found that further deepening the network beyond 20 layers led to overfitting, with peak accuracy observed at VGG-20. The ensemble of AlexNet and VGG-19 reduced false negatives by 36% relative to individual models, achieving the highest overall accuracy of 99.37% with a threat detection rate of 99.06% [7].

This study makes a critical practical observation: toy guns, which have shapes similar to real firearms but differ in material composition, introduced valuable material variation into the training set. This suggests that material-based features play a distinct and complementary role to shape-based features — a principle that directly motivates the two-model property-extraction architecture proposed in this project.

3.2 Transfer Learning for X-Ray Classification

Akçay et al. (2016) demonstrated the viability of transfer learning for X-ray baggage security screening. Pre-trained CNN models, initially trained on large-scale natural image datasets such as ImageNet, were fine-tuned on X-ray security imagery. Their experiments showed that transfer learning significantly reduces the need for large domain-specific labeled datasets, which are particularly scarce in the security domain due to classification restrictions on threat imagery [10].

The broader significance of this line of research lies in its implication for scalable system design. If pre-trained feature extractors can be adapted to X-ray imagery with minimal retraining, then the property extraction model in a two-stage system can leverage general visual representations while being fine-tuned for X-ray-specific material and shape properties.

3.3 Multi-Scale CNN for Material Classification

Benedykciuk et al. (2021), writing in *Signal, Image and Video Processing* (Springer), proposed a multi-scale Convolutional Neural Network for material classification in X-ray security images. Their system classified object patches into categories including organic, metallic, and composite materials, by processing multiple patch sizes in parallel. This multi-scale approach captures both fine-grained texture information and broader contextual features simultaneously [8].

The paper identifies a core challenge that is directly relevant to this project: materials that are visually similar in shape may be entirely distinct in composition, and vice versa. This property makes pure shape-based detection unreliable for certain threat categories. The multi-scale material classifier proposed by Benedykciuk et al. serves as a strong precedent for the Material Property Extraction component of the proposed architecture.

4. Real-Time Object Detection Using YOLO Architectures

While CNN classifiers can determine whether a threat object is present in an image, they are insufficient for real-time deployment because they cannot simultaneously localize the object within the image. Object detection architectures, which combine classification and spatial localization into a unified framework, are therefore essential for operational security screening systems.

4.1 YOLOv2, YOLOv3, and Tiny-YOLO for Baggage Screening

Deshpande et al. (2022) evaluated three YOLO-family models — YOLOv2, YOLOv3 (fixed resolution), and YOLOv3 with multiscale training — for automated threat detection in baggage images. YOLOv3 with multiscale training at resolutions ranging from 320x320 to 608x608 pixels achieved a mean Average Precision (mAP) of 99.17%, an accuracy of 98.18%, and an F1 score of 98.17% [7]. For non-GPU environments, Tiny-YOLOv3 processed images in 0.369 seconds on CPU — within the 2.5-second display window for standard X-ray inspection conveyors — representing the best available trade-off between speed and accuracy.

This work, conducted on the ECIL ECX6040:SVXBIS dual-energy scanner, established YOLO-based detection as not merely an academic exercise but a deployment-ready solution integrated as a pluggable add-on to existing XBIS software infrastructure. This real-world integration is significant because it validates that AI-based detection can be embedded non-disruptively into existing airport security workflows.

4.2 Improved YOLO Variants for X-Ray Security

A substantial body of research has focused on adapting and improving YOLO architectures specifically for X-ray security imagery. Park et al. (2025), in *Radiation Physics and Chemistry* (Elsevier), developed a YOLO-based CNN object detection model using a CSP-Darknet53 backbone trained on 6,000 annotated X-ray images. The model achieved an inference time of 12.58 milliseconds per baggage item and effectively handled overlapping objects through multi-stage occlusion management [6].

Jing et al. (2023), in a study published in MDPI Sensors, proposed EM-YOLOv7 — a variant that fuses an Edge Feature Extractor (EFE) with a Material Feature Extractor (MFE). This dual-stream design is particularly noteworthy in the context of this project because it mirrors the fundamental insight behind property-based detection: shape (edge) features and material features should be extracted and fused explicitly, rather than relying on a single network to implicitly learn both [14].

Miao et al. (2019), in a landmark paper at CVPR, introduced the SIXray dataset — a large-scale security inspection X-ray benchmark comprising over one million images with six categories of prohibited items. They proposed a Class-Balanced Hierarchical Refinement (CHR) network to address the severe class imbalance inherent in real-world X-ray screening datasets [15]. The SIXray dataset has since become the most widely used benchmark for evaluating X-ray detection models, and several works reviewed here report results on this dataset.

4.3 Lightweight and Deployable YOLO Models

Zhang et al. (2024), in ScienceDirect, proposed the Tinyray model — an improved YOLOv7-tiny architecture using FasterNet as the backbone and Partial Convolution (PConv) modules. The model achieved 88.7% identification accuracy with a detection speed of 89.5 FPS, representing a 14.7% reduction in model parameters compared to baseline [16].

Zhou et al. (2024) introduced Fine-YOLO, which uses a Low-Parameter High-Density Feature Integration module to improve small target detection in X-ray images. Their model was evaluated on the EDS dataset — a benchmark covering 10 prohibited item categories from three different scanner manufacturers — achieving strong generalization across scanner types [18].

ScanGuard-YOLO (2024), evaluated on OPIXray and HiXray datasets, improved recall by 4.2% and F1 score by 2.4% over the YOLOv8s baseline by introducing dilated convolutions and RepGFPN neck modules to increase receptive field size and capture global features more effectively [17].

5. Dataset Construction and Threat Image Projection

A recurrent challenge in the X-ray security domain is the extreme scarcity of labeled threat images. Unlike general-purpose computer vision datasets, threat imagery carries legal and security restrictions that limit data collection and sharing. Models trained on imbalanced datasets — where non-threat images vastly outnumber threat images — tend to develop a strong bias toward the negative class, resulting in unacceptably high false-negative rates.

Rogers et al. (2016) introduced the Threat Image Projection (TIP) technique as a principled solution to this data scarcity problem. TIP works by computationally compositing real threat object images onto clean baggage scans using the Beer-Lambert law, which models photon attenuation through materials. A threat mask is computed based on the attenuation of the background image, and the threat image is projected into the clean bag image by pixel-wise multiplication [1].

Deshpande et al. (2022) applied TIP to generate a training set of approximately 1.24 lakh (124,000) augmented images from a base set of only 80 non-threat and 73 threat images. Class-preserving translation and rotation augmentations were further applied to expose the model to natural variations in object orientation and position [7]. This augmentation pipeline is a direct methodological precedent for any future dataset construction effort in the proposed project.

The broader significance of TIP is that it enables the systematic generation of diverse, balanced training data without requiring physical access to actual threat objects. For deployment in green channel screening — where the baseline assumption is that passengers carry no restricted items — TIP provides the only practical means of generating sufficient threat-positive examples to train a reliable detection model.

6. Property-Based and Attribute-Driven Object Detection

The conventional object detection pipeline treats the problem as end-to-end classification: given an input image, predict bounding boxes and class labels directly. While this approach has achieved remarkable results on general-purpose datasets, it has significant limitations in the security screening domain. The set of potential threat objects is open-ended and constantly evolving, meaning models trained on fixed class sets become obsolete as new threats emerge. Furthermore, direct class-label predictions offer limited interpretability — a critical deficiency in safety-critical applications where human operators must trust and verify model outputs.

A property-based detection paradigm addresses both limitations by decomposing the detection task into two stages: (i) extracting physical and material properties of detected objects, and (ii) classifying objects based on those properties. This two-stage approach is more scalable because new object categories can be introduced by defining new property patterns, without retraining the feature extraction model. It also improves interpretability because the system can explain its classifications in human-readable terms, such as 'high-density metal, sharp edges, blade-like geometry.'

Rogers et al. (2017) explored a deep learning framework for the automated inspection of complex dual-energy X-ray cargo imagery, utilizing anomaly detection techniques that move beyond fixed-class recognition. Their work demonstrated that systems capable of detecting unusual material compositions or structural anomalies — rather than matching against a fixed object class catalog — are more robust to novel threat items [4].

Akçay and Breckon (2022), in a comprehensive survey published in *Pattern Recognition* (Elsevier), reviewed the spectrum of deep learning advances in X-ray security imaging. The survey highlights a growing research direction toward unsupervised and semi-supervised anomaly detection, which aligns closely with the property-based philosophy. They identify that one of the key open problems in the field is developing systems that can generalize to new threat types without full retraining [10].

The color-based segmentation approach proposed by authors in ResearchGate (2020), which segments X-ray images into organic, inorganic, mixed, and opaque material regions before classification, provides a practical precedent for the material property extraction stage in the proposed architecture. Their ML-based pipeline demonstrated that color-coded material regions can be reliably segmented and used as the basis for downstream object identification [5].

7. Handling Occlusion and Segmentation in Baggage X-Ray Images

One of the most persistently challenging aspects of X-ray baggage analysis is the problem of object occlusion. Unlike natural scene images, where objects typically appear as discrete entities, luggage contents are densely packed. Multiple objects overlap, creating composite image regions that confound both human operators and automated detection systems.

Zhao et al. (2022), in *IEEE Transactions on Information Forensics and Security*, proposed a Label-Aware Mechanism (LAM) to detect overlapping objects in X-ray security imagery. Their method uses label information to guide attention toward occluded object regions, improving detection accuracy in scenarios with heavy clutter [19].

Wei et al. (2020), in a paper presented at ACM Multimedia, introduced the OPIXray dataset — the first X-ray security benchmark specifically designed for occluded prohibited item detection — alongside a De-occlusion Attention Module (DOAM) that suppresses background clutter and focuses on occluded regions of interest [20].

Shao et al. (2022) exploited foreground and background separation for prohibited item detection in overlapping X-ray images, using a two-branch network that processes foreground objects and background context separately before fusing their representations [21]. This separation strategy is directly relevant to the proposed property extraction model, which must isolate individual objects before computing shape and material features.

These works collectively demonstrate that robust segmentation — the ability to isolate and delineate individual objects within cluttered X-ray images — is a prerequisite for any downstream property extraction process. The proposed architecture's pre-processing pipeline must therefore incorporate advanced segmentation techniques to ensure that Model 1's property extraction operates on well-isolated object regions.

8. Benchmark Datasets for X-Ray Security Research

The availability of standardized, publicly accessible datasets has been a critical enabler of reproducible research in X-ray security detection. Several major benchmark datasets have been established that define the evaluation landscape for this domain.

The SIXray dataset (Miao et al., CVPR 2019) is the largest publicly available X-ray security benchmark, containing 1,059,231 images in six categories: gun, knife, wrench, pliers, scissors, and hammer. Approximately 8,929 images contain prohibited items, representing a class imbalance ratio of roughly 1:117. The dataset introduced challenging scenarios involving heavily overlapping objects in real airport X-ray images [15].

The PIDray dataset (Zhang et al., 2023, International Journal of Computer Vision) is a large-scale benchmark with 47,677 images spanning 15 prohibited item categories, designed to reflect realistic prohibited item detection challenges including concealment scenarios [21].

The HiXray dataset was collected from real daily security checks at international airports, with images manually labeled by professional screeners. It includes 45,364 images across 8 categories and is considered one of the most representative of real-world operational conditions [17].

The OPIXray dataset (Wei et al., ACM Multimedia 2020) provides 8,885 X-ray images of cutting instruments across five sub-categories, with varying degrees of occlusion, specifically designed to benchmark performance under overlapping object conditions [20].

For this project, the ECIL ECX6040:SVXBIS scanner images used by Deshpande et al. represent a real-world Indian operational context. Future research in this domain should prioritize creating or contributing to publicly available dual-energy X-ray datasets representative of South Asian airport security environments.

9. Comparative Analysis of Existing Approaches

The following table summarizes the key attributes of representative works reviewed in this survey, organized by their methodological approach and key performance characteristics.

Study / Author	Approach	Model / Method	Key Metric	Limitation Noted
Deshpande et al. (2022)	CNN Classification + YOLO Detection	VGG-19, AlexNet Ensemble; YOLOv3 Multiscale	mAP 99.17%; Accuracy 99.37%	Fixed class labels; limited scalability
Benedykciuk et al. (2021)	Material Classification	Multi-Scale CNN (6 material classes)	High material classification accuracy	Fixed 6 classes; no object localization
Jing et al. (2023)	Edge + Material Fusion Detection	EM-YOLOv7 with EFE + MFE modules	Strong SIXray benchmark performance	Limited to predefined classes
Park et al. (2025)	Real-Time CNN Detection	YOLO with CSP-Darknet53 backbone	12.58 ms inference time	Non-rigid contraband not handled
Zhao et al. (2022)	Overlapping Object Detection	Label-Aware Mechanism (LAM)	Improved detection under occlusion	Complex label annotation required
Miao et al. (2019)	Large-Scale Benchmark + CHR Network	Class-Balanced Hierarchical Refinement	SIXray benchmark standard	Dataset bias toward specific threat types
Proposed Architecture	Property-Based Two-Model Detection	Model 1: Feature Extractor; Model 2: Property Classifier	Scalable; Interpretable; Adaptable	Two-model pipeline overhead; requires precise segmentation

10. Research Gaps and Motivation for the Proposed Work

The survey of existing literature reveals several critical research gaps that motivate the proposed property-based detection architecture:

1. **Scalability to Novel Threats:** Existing deep learning models are trained on fixed class sets. When new prohibited items emerge, the entire model pipeline must be retrained on new labeled data. The property-based approach decouples feature extraction from object

identification, allowing new threat categories to be defined through property patterns without full retraining.

2. **Interpretability Deficit:** Current state-of-the-art detection models function as black boxes. They produce class predictions without human-understandable explanations, limiting operator trust and accountability. Property-based systems generate structured descriptions of why an object was flagged, addressing a fundamental requirement in safety-critical deployment contexts.
3. **Dataset Dependency:** Large-scale labeled datasets for every possible threat category are unavailable and impractical to assemble. The property-based architecture reduces dependency on massive per-class labeled datasets by training the property extractor once on general object features and adapting the identification model through property definitions.
4. **Green Channel Deployment Gap:** Most existing research focuses on high-risk screening lanes. The specific operational context of green channels — where the prior probability of threat items is very low — introduces distinct false-positive management challenges that are not adequately addressed in the literature.
5. **Material-Shape Integration:** While some works like EM-YOLOv7 fuse edge and material features, there is limited research on explicitly structured, property-vector-based identification pipelines that treat shape and material as independently computed, structured feature spaces. The proposed dual-model architecture contributes directly to this underexplored paradigm.

11. Conclusion

This literature survey has reviewed the foundational and state-of-the-art research underpinning the proposed AI/ML-Based Object Detection System for Dual-Energy X-Ray Luggage Scanners. The review spans dual-energy X-ray imaging physics, CNN-based classification and VGGNet ensemble architectures, real-time YOLO-family detection models, material classification networks, dataset construction through Threat Image Projection, and property-based detection paradigms.

The surveyed literature collectively demonstrates that while deep learning has achieved near-human performance on fixed-category threat detection benchmarks, critical gaps remain in scalability, interpretability, and adaptability to emerging threats. The proposed two-model property-extraction architecture addresses these gaps by separating the concerns of visual feature extraction from semantic object identification, enabling a more flexible and transparent detection pipeline.

The strongest conceptual precedents for the proposed work are the material classification approach of Benedykciuk et al. (2021), the dual-stream edge-and-material fusion of EM-YOLOv7 by Jing et al. (2023), and the anomaly detection direction identified by Akcay and Breckon (2022). Building on these foundations, this project proposes to operationalize the property-based paradigm into a fully integrated, deployable system for dual-energy X-ray airport baggage screening.

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